Clinical Pharmacogenetics Pipeline

Background

We are integrating a Pharmacogenetics (PGx) feature into our existing clinical NGS platform. The goal is to process WGS/WES data to predict drug response phenotypes for clinically relevant targets.

Part 1: The Technical Implementation (Code)

Develop a bioinformatics pipeline using **Nextflow** and **Python**.

- **Input:** An aligned BAM/CRAM file (hg38).
- **Output:** A structured JSON file containing:
    1. Identified alleles (star-allele nomenclature, e.g., *1/*17)
    2. Predicted phenotype (e.g., “Rapid Metabolizer”)
    3. A confidence score or quality metric for the call

Constraints:

- The workflow must utilize **Docker** containers.
- The parsing and logic layer must be written in **Python**.
- The code must be production-ready (modular, with error handling and type annotations).

Part 2: The Presentation (Feasibility & Strategy)

Prepare a 20-minute presentation (slides) to be delivered to the Product teams during the next interview round.

Agenda for the Presentation:

1. **Current state of the art**
    - How do other companies approach pharmacogenomics analysis? Please provide examples of how results are typically reported?
    - What is the general trend regarding the use of pharmacogenomics in Rare Disease patients? Is it currently standard clinical practice, considered innovative, or strictly research-based? Is it clinically relevant for this cohort?
    - Discuss the trade-offs between using WES, WGS, Short-read, and Long-read sequencing for PGx. Which is the industry moving towards?
2. **Architectural Defense:**
    - Walk us through your tool selection and pipeline design.
    - Why did you choose specific algorithms/tools?
    - How does your pipeline handle the specific complexities of difficult genes (e.g., *CYP2D6* pseudogenes)?

3. **Feasibility Analysis:**
    - Discuss the technical limitations of using standard NGS (WES/WGS) data for Pharmacogenetics.
    - How would you validate this pipeline clinically before releasing it?
4. **Product & UI/UX Vision:**
    - A physician is about to prescribe a drug and needs to know whether the patient can metabolize it. Present a visual concept (wireframe or mockup) of how this data should appear in a clinical interface.
    - Demonstrate how you would communicate “uncertainty” or “complex genotypes” to a clinician without overwhelming them.